

SUSHANT LOKHANDE

San Jose, CA | +1 669-340-6655 | lokhandesushant014@gmail.com | sushantlokhande.me |
linkedin.com/in/sushantlokhande14 | github.com/sushantlokhande14

EXPERIENCE

San Jose State University | Graduate Research Assistant, Computer Vision & ML Aug 2025 to May 2026

- Boosted malware-family classification to **80.2%** accuracy across **17 families** by soft-voting **ResNet50**, **ViT-B/16**, VGG16, HOG, and Haralick models, beating the best single model by **2.4 pts** (McNemar $p < 0.001$).
- Cut experiment setup effort by about **55%** by automating orchestration, configuration, **telemetry** collection, and analysis, making large experiment sweeps reproducible.
- Reduced rebuffering by **19%** versus a fixed-rate baseline by building a **Docker**, **WebTransport**, and **Media-over-QUIC** testbed and running controlled adaptation experiments across network and protocol configurations.
- Built and evaluated **RAG** pipelines (chunking, embeddings, vector retrieval, top-k, grounded prompts), separating **retrieval** failures from generation failures to guide fixes.

Turito | Software Engineering Intern, Real-Time Video Systems Jun 2022 to Oct 2022

- Slashed p95 **CPU** by **22%** by profiling high-frequency media paths and removing excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Cut controlled-disruption failures by **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Worked across a **WebRTC** control plane and **RTP/RTCP** media paths spanning about **12 regions**, including ICE/STUN/TURN connection setup and degraded-network handling.
- Reduced production diagnosis time by **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Surfaced backend fixes that raised equivalent-load signaling throughput by about **31%**.

TECHNICAL SKILLS

Languages: Python, C++17, Rust, SQL, TypeScript

Computer Vision & ML: PyTorch, OpenCV, ViT, ResNet, EfficientNet, Siamese networks, triplet loss, HOG, Gabor, scikit-learn

Video & Systems: WebRTC, RTP/RTCP, QUIC, AVX2/SIMD, profiling, multithreading, Linux

Infra: Docker, FastAPI, gRPC, Kafka, PostgreSQL, Redis, AWS

PROJECTS

Contactless Palmprint Verification | *PyTorch, OpenCV, Siamese networks*

- Fused EfficientNet, ResNet, ViT, LBP, KAZE, Gabor, DWT, and DCT features into a 256-d triplet-loss embedding reaching **0.9975 AUC** and **2.13% EER**.

Proxima, Vector Search Engine | *C++17, AVX2/SIMD, HNSW, pybind11*

- Built an HNSW vector search engine from scratch that reaches **0.999 recall** at **12,049 QPS** single-threaded, **1.8x** faster than hnswlib and **2.5x** faster than FAISS on 100K vectors.

OPEN SOURCE

- **NVIDIA/garak**: preserved attempt UUIDs in report aggregation for traceable LLM evals (merged, shipped in v0.17.0)
- **fission/fission**: 3 merged PRs, Kubernetes event and KEDA object capture in support dumps, removed duplicate package enumeration

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Machine Learning, Biometric Security, GPU Programming, Artificial Intelligence, Distributed Computing, Advanced Algorithms

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)